Small Exchange FIX Order Management API Specification

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1 Revision History

Version	Date	Author	Description
0.1	Nov-10-2017	Sergey Samushin	Initial draft
0.2	Dec-20-2017	Sergey Samushin	- Added multi-leg cancel/replace message specification - Added “Leg Instrument” component specification, multi-leg messages changed accordingly - Added LegSide and LegRatioQty for multi-leg order requests - Multi-leg order fills reported for legs individually, added note to Trade Execution report message
0.3	Mar-29-2018	Sergey Samushin	- Added IOC and FOK time-in-force qualifiers support - Added MinQuantity and PositionEffect order fields - Changed instrument identity fields. Instrument is identified by Symbol<55> or LegSymbol<600> tags
0.4	May-7-2018	Sergey Samushin	- Added SenderSubID and TargetSubID standard header tags. Changed description for SenderCompID and TargetCompID tags. - Added OrigSendingTime standard header tag - Cancel and Order status messages require providing OrderID<37> Exchange order ID.
0.5	June-1-2018	Sergey Samushin	- Added BusinessMessageReject message - Added Text tag with max length of 120 to OrderCancelReject message
0.6	Aug-24-2018	Sergey Samushin	Described contract of Side and Price fields for multi leg orders
0.7	Oct-15-2018	Sergey Samushin	- Cancel request requires Instrument Component to be specified. OrderId<37> is optional. - Ability to send multi-leg orders with Sell side.
0.8	Dec-17-2018	Sergey Samushin	Added Exec report “L” for triggered stops
0.9	Jan-24-2019	Yury Kudryashov	- Exec reports for expired and done-for-day orders - Details on order identifiers - Reset schedules - Description of firm identifiers - Details on trade bust/correct execution reports
0.10	Jan-25-2019	Yury Kudryashov	- CTI codes (tag 9702) in incoming orders and execution reports - Customer or Firm (tag 204) in incoming orders and execution reports

			- Self-match prevention described (tags 7928, 8000)
0.11	Mar-4-2019	Yury Kudryashov	- Wording and samples fixes - Length of firm identifiers and self-match tokens clarified
0.12	Mar-21-2019	Yury Kudryashov	- MinQty (110) tag in execution reports - SelfMatchPreventionToken (7928) replaced with SelfMatchPreventionID (2362) - Standard tags 528 and 582 are used to collect CTI codes and customer/firm order origin - ManualOrderIndicator (1028) tag - Parties component
0.13	Aug-05-2019	Yury Kudryashov	- LegRefID (654) in trade execution reports for spread fills - PartyRole (44), ExecID (17), OrderID (37), SolicitedFlag (377) values clarification - Subscriber token
0.14	Nov-18-2019	Yury Kudryashov	- Subscriber token length clarified (15 characters) - Logon and sequence reset protocol revisited and detailed (Logon Scenarios and Sequence Reset) - Order types / TIF compatibility matrix (Supported Order Types and TIF) - Example of a multi-leg order fill - Position effect / leg position effect clarification - Stop order behavior clarification - Mutiple wording and typo fixes - Confidentiality / disclaimer
0.15	Nov-29-2019	Irina Timkina	TrdMatchID(880), AggressorIndicator(1057) in trade execution reports
0.16	Jan-01-2020	Yury Kudryashov	Logon and sequence reset protocol revised and detailed again
0.17	Feb-14-2020	Vladimir Danilov	Account field length changed to 10 characters
1.0	Mar-03-2020	Yury Kudryashov	Tag 369 (LastMsgSeqNumProcessed) described
1.1	May-13-2020	Vladimir Danilov	SelfMatchPreventionId(2362) field length changed to 15 characters
1.2	June-18-2020	Natalie Oblazny Yury Kudryashov	- Clarifies purpose and scope of tag 528 and tag 582 - MinQty (110) value for IOC orders should be > 0 - Maximum supported sequence clarified - Clarification on multi-leg orders prices and sides
2.0	Sep-22-2020	Sergey Samushin	Options trading support
2.1	Oct-05-2020	Sergey Samushin	Updated FIX connection optional configuration features: cancel on disconnect, default self-match prevention, timestamps microseconds format
2.2	Dec-31-2020	Sergey Samushin	Added Mass Quote messages

2.3	Feb-5-2021	Vladimir Danilov	Added details on throttling limits
2.4	Mar-4-2021	Sergey Samushin	Added Mass Quoting constraints and behavior details
2.5	Mar-19-2021	Nikita Duginets	Added reject codes for Mass Quote acknowledgment
2.6	Mar-30-2021	Sergey Samushin	Added 304 (TotNoQuoteEntries) mandatory for Mass Quote message.
2.7	Jul-2-2021	Sergey Samushin	Added supported order types matrix for futures/options multi-leg orders.

2 Confidentiality/Disclaimer

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3 Overview

This document describes the Small Exchange's order management FIX API for firms connecting directly to the Small Exchange. The API allows connected firms to send, modify and cancel their orders.

The API uses FIX protocol version 4.4. This document describes the messages supported by the Small Exchange.

Note that FIX tags other than supported by the Exchange specification are ignored and not processed.

The document is not intended to serve as a full-fledged specification of the FIX protocol. It is assumed the reader is acquainted with the protocol, please refer to the official FIX specifications at https://www.fixtrading.org/ website for additional details.

3.1 The Small Exchange Hours of Operation

The Small Exchange's current trading schedule can be found at http://smallexchange.com/. Orders entered outside trading hours will be rejected. Firms are encouraged to stay connected 15 minutes after the official close time to receive reports that are generated after trading session closing logic is run (e.g. Done for Day or Expired reports).

3.2 Weekly Sequence Reset

The Exchange will reset its FIX sequences on a weekly basis during the weekend. See details in Logon Scenarios and Sequence Reset.

3.3 Certification

To be able to submit orders to the Small Exchange, direct market access firms must be certified. The Small Exchange provides a separate environment for integration, acceptance testing and certification. Please contact the Small Exchange Infrastructure team at it-ops@smallexchange.com to obtain additional information.

3.4 Firm Identifiers

All messages sent from the Exchange and to the Exchange *must* contain both the SenderCompID (49) and TargetCompID (56) fields. The firm and the Small Exchange agree on these values at the time of onboarding.

Field	Firm to Exchange	Exchange to Firm
SenderCompID	ID of the connection. A firm may have multiple connections, each connection will have its own **CompID**	Always “SMALLEX”
TargetCompID	Always “SMALLEX”	ID of the connection. A firm may have multiple connections, each connection will have its own **CompID**

Maximum length of a connection identifier is 32 characters.

All application messages sent to or from the Small Exchange *must* also contain both the SenderSubID (50) and TargetSubID (57) fields.

Field	Firm to Exchange	Exchange to Firm
SenderSubID	Unique trading firm code as assigned by the Exchange during initial setup.	Identifies the environment on the Exchange side. This can be either “PROD” or “TEST”. The firm can use this value to guard itself from accidentally issuing UAT or test orders on the live market.
TargetSubID	Identifies the target environment on the Exchange side. This can be either “PROD” or “TEST”. The firm can use this value to guard itself from accidentally issuing UAT or test orders on the live market. The Exchange will reject any messages sent to unexpected environment (e.g. production environment will reject any TEST messages).	Contains the unique trading firm code as assigned by the Exchange during initial setup.

Maximum length of a firm identifier is 4 characters.

3.5 Order Identifiers

All orders submitted to the Small Exchange must have unique ClOrdID (tag 11).
The Exchange enforces the uniqueness of the identifier among all working orders of a trading firm across all its FIX connections. It is expected that trading firm keeps ClOrdID tag unique across all its FIX connections to avoid referencing same orders sent from different FIX connections, and issues with reporting, clearing, and support.

ClOrdID can be at most 20 characters long.

3.6 Operator Identification

All orders submitted to the Small Exchange must contain identification of the operator who placed the order. This is conveyed via the Parties FIX component with PartyRole = 44 (see PartyRole (tag 452). See more details in Parties Component description.

3.7 Exchange Subscriber Identification

Orders submitted to the Small Exchange by the subscriber can contain a token issued by the Exchange to the subscriber. The token is captured by the Exchange and is echoed back in execution reports. This is conveyed via the Parties FIX component with PartyRole = 2101 (see PartyRole (tag 452). See more details in Parties Component description.

Maximum length of the subscriber token is 15 characters.

3.8 Trade Busts and Corrects

The Small Exchange FIX Order Management API does **not** distribute unsolicited reports about trade busts and corrects. The firms are expected to utilize the FIX Drop Copy connections to receive these messages.

3.9 Done for Day and Expiration Reports

After the trading session is closed FIX connection distributes two types of events designating status of trading firm working orders:

- Execution Report: Expired (MsgType = 8, ExecType = C) is sent for all DAY orders expired due to trading session closing. The report is also sent for GTC orders on their instrument last trading day session closing.
- Execution Report: Done for Day (MsgType = 8, ExecType = 3) is sent for all GTC orders working and restated for the next trading day. Trading firms don't need to send any instructions for GTC orders restatement, this is conducted by the Exchange automatically.

3.10 Self-Match Prevention

Self-match prevention functionality allows market participants to prevent Buy and Sell orders for the same account, firm or group of accounts to match with each other. The functionality is optional and is controlled with a pair of FIX tags in incoming orders:

- **SelfMatchPreventionID** (2362) – orders with the same Self Match Prevention ID for the same executing firm will not match. Maximum length of the ID is 15 characters.
- **SelfMatchPreventionStrategy** (8000) – this value defines the strategy of dealing with matching orders if self-match prevention is triggered. The Exchange will either cancel the aggressor order (unsolicited), the resting order or both. The Exchange uses the strategy from the *aggressor* order to deal with self-matched orders.

The API will send the actual values of SelfMatchPreventionID and SelfMatchPreventionStrategy in all execution reports.

FIX connection of a trading firm can be configured with a default pair of self-match ID and strategy to be used for all orders received via the connection. Contact it-ops@smallexchange.com to request the configuration.

3.11 Supported Order Types and TIF

During regular trading sessions the following order types / TIF combinations are supported:

	DAY	GTC	IOC	FOK
Limit	Supported	Supported	Supported	Supported
Market	Supported	Not Supported	Not Supported	Not Supported
Stop	Supported	Supported	Not Supported	Not Supported
Stop-Limit	Supported	Supported	Not Supported	Not Supported

During the auction (pre-open / pre-open no cancel) sessions the following order types / TIF combinations are supported:

	DAY	GTC	IOC	FOK
Limit	Supported	Supported	Not Supported	Not Supported
Market	Not Supported	Not Supported	Not Supported	Not Supported
Stop	Supported	Supported	Not Supported	Not Supported
Stop-Limit	Supported	Supported	Not Supported	Not Supported

Unsupported order type / TIF combinations are rejected.

Supported order types also depend on order instrument:

	Single Futures	Multi-leg Futures	Single Option	Multi-leg Option
Limit	Supported	Supported	Supported	Supported
Market	Supported	Supported	Supported	Not Supported
Stop	Supported	Supported	Supported	Not Supported
Stop-Limit	Supported	Supported	Supported	Not Supported

Multi-leg Futures are two-legged spread orders. For instance, Futures Calendar spread.
Multi-leg Option orders have at least one option leg. Covered Call strategy is treated as multi-leg option.

3.12 Timestamps Precision

FIX connection can be configured for a trading firm to support microseconds precision and **YYYYMMDD-HH:MM:SS.ssssss** format for UTCTimestamp type tags (for instance, SendingTime<52>, TransactTime<60>, OrigSendingTime<122>) in inbound and outbound messages. By default FIX connection is configured with standard milliseconds precision for UTCTimestamp tags. Contact it-ops@smallexchange.com to request the configuration.

3.13 Cancel on Disconnect

Automatic orders cancellation for working orders sent within a FIX connection can be optionally configured for a FIX connection to cancel only DAY or ALL (DAY and GTC) orders on FIX client disconnect. Note, that the operation of orders cancelling is not performed on FIX API disconnect when instrument market is closed, paused or halted.

3.14 Message Throttling

The Exchange configures a FIX connection to limit rate messages sent by participants. A limitation on the number of messages by type sent over a time interval can be imposed (msg/sec):

- New and Cancel/Replace Order Messages
- Cancel Order Messages
- Mass Quote Messages
- Quote Cancel Messages

If count of the messages for an interval is exceeded then further messages are not processed by the Exchange and rejected with a Business Reject <MsgType = j> message. The BusinessRejectReason<380> field is set to “8” (THROTTLE_LIMIT_EXCEEDED) and Text<50> field is set to “Throttling limit (limit=<Limit>;interval=<Interval>;penalty=<Penalty>)” with exact values of configured limit, interval and penalty as time left till the interval end.

Example: throttling reject for a New Order Single <MsgType = D>:

MsgType = j (*from client to the Exchange*)
RefMsgType = D (*answer for New Order Single*)
BusinessRejectReason = 8 (*Throttling limit exceeded*)
Text=Throttling limit exceeded (limit=200;interval=1000ms;penalty=300ms)
(limit of 200 orders/second expires in 300 ms)

Contact it-ops@smallexchange.com to request current FIX throttling limits and intervals configuration.

4 Session Protocol

Session protocol assures client identification, sequential request processing, session state control and ability to restore the session after downtime. In the scope of a session, all FIX messages are identified by unique integer sequence numbers and are processed in that order. When an incoming sequence number does not match the expected one, the session must be recovered. If incoming sequence number is less than expected and PossDupFlag is not set to Y, then it is considered a fatal error, and the connection is dropped by the server. If the sequence number of incoming message is greater than the next expected number, then the Resend Request is issued for missed messages.

Client and server use the following administrative messages:

- **Logon –** initiates (client) or approves (server) session opening.
- **Logout –** initiates or approves session closing.
- **Resend Request –** requests missed FIX messages.
- **Sequence Reset – Gap Fill**, must be used instead of resend of administrative messages.
- **Test Request –** used to control the session state. Requires a reply **Heartbeat** message with properly filled TestReqID (112) field.
- **Heartbeat –** used to control session connection state.
- **Reject –** for administrative message reject.

Maximum sequence number supported by the Exchange is 9223372036854775807 ($2^{63} – 1$).

4.1 Logon Scenarios and Sequence Reset

The Exchange will reset both incoming and outgoing message sequence numbers to 1 during the weekend. Note that the use of the ResetSeqNum (141) tag can result in message loss and should generally be avoided. The Exchange does not require the use of this flag under normal operating conditions.

For the **first logon** on any week the Logon message can have its MsgSeqNum (34) set to 1 or the sequence can continue. The Exchange will reset the sequences if incoming MsqSeqNum (34) is greater than 1. If there are any undelivered messages for this connection, the Exchange will resend them after a successful logon.

For a **mid-week logon** (second and subsequent logon attempts on any week) the Logon message sequence should continue from the next sequence number where the client logged out or disconnected. If there are any undelivered messages for this connection, the Exchange will resend them after a successful logon.

If an incoming Logon message (first or midweek) has ResetSeqNum (141) flag set to ‘Y’, the Exchange will discard all undelivered messages it might have queued for this connection and reply with MsqSeqNum (34) = 1.

5 Message Format

All FIX messages, either administrative or business, require standard header and trailer components. Note that SenderCompID and TargetCompID values pair is constant for a single FIX connection between a client and the Small Exchange system (see Firm Identifiers).

5.1 Standard Header Component

Tag	Field Name	Type	Req	Comments
8	BeginString	String	Y	"FIX.4.4"
9	BodyLength	Length	Y	Message length excluding the CheckSum field
35	MsgType	String	Y	Message type
49	SenderCompID	String	Y	Connection identifier assigned by the Small Exchange. "SMALLEX" for messages sent from the Exchange system. See Firm Identifiers.
50	SenderSubID	String	Y	For messages sent to the Exchange contains the trading firm code. For messages from the Exchange it is an identifier of the message originating system ("PROD" or "TEST"). See Firm Identifiers.
56	TargetCompID	String	Y	"SMALLEX" for messages sent to the Exchange. Connection identifier assigned by the Small Exchange for messages sent from the Exchange system. See Firm Identifiers.
57	TargetSubID	String	Y	For messages sent to the Exchange identifies destination system – "PROD" or "TEST". For messages sent from the Exchange contains the trading firm code. See Firm Identifiers.
34	MsgSeqNum	SeqNum	Y	Integer message sequence number
43	PossDupFlag	Boolean	C	The Exchange ignores this field on all incoming messages and relies on ClOrdID uniqueness instead.
52	SendingTime	UTC Timestamp	Y	Time of message transmission (always expressed in UTC)
97	PossResend	Boolean	C	The Exchange ignores this field on all incoming messages and relies on ClOrdID uniqueness instead.
122	OrigSendingTime	UTC Timestamp	C	Original time of message transmission, required for messages resent as a result of a ResendRequest
369	LastMsgSeqNum Processed	SeqNum	N	Last incoming sequence received and processed by the Exchange (validated and persisted, but not

				necessarily sent to the downstream matching engine). The Exchange sends this field in all outgoing messages and ignores this value in incoming messages.

5.2 Standard Trailer Component

Tag	Field Name		Req	Comments
10	CheckSum	String	Y	Three byte, simple checksum. Always defined as three characters

6 Administrative Messages

6.1 Logon (MsgType = A)

Initiates connection from client side and approves connection if sent by the Exchange. ResetSeqNumFlag allows clients to start a new session (reset session sequence numbers). See Logon Scenarios and Sequence Reset.

The Heartbeat Interval is declared by the session initiator using the HeartBtInt field in the Logon message. The heartbeat interval timer should be reset after every message is transmitted (not just heartbeats). The HeartBtInt value should be agreed upon and specified by the Logon initiator (client) and echoed back by the Logon acceptor. The same HeartBtInt value is used by both sides, the Logon "initiator" and Logon "acceptor".

Tag	Field Name	Type	Req	Comments
<Standard Header>			Y	MsgType = "A"
98	EncryptMethod	Int	Y	"0" - None. Security must be guaranteed on transport level
108	HeartBtInt	Int	Y	Heartbeat interval in seconds
141	ResetSeqNumFlag	Boolean	N	"N" = use previous sequences "Y" = reset sequences (start a new session). The use of the ResetSeqNum (141) tag can result in message loss and should generally be avoided. The Exchange does not require the use of this flag under normal operating conditions. See Logon Scenarios and Sequence Reset. If clients cannot recover the previous session they start new session with 1 and set this field to "Y".
<Standard Trailer>			Y	

Example: logon with MsgSeqNum too low:
MsgType = A (*from client to the Exchange*)
SenderCompID = dx-fix-trade1p
TargetCompID = SMALLEX
SenderSubID = DX
TargetSubID = TEST
MsgSeqNum = 1
SendingTime = 20191120-15:01:53

MsgType = A (*from the Exchange to client*)
SenderCompID = SMALLEX
TargetCompID = dx-fix-trade1p
SenderSubID = TEST
TargetSubID = DX
MsgSeqNum = 3
SendingTime = 20191120-15:01:53
Text = MsgSeqNum too low, expecting 3 but received 1

6.2 Logout (MsgType = 5)

The message initiates or confirms termination of a FIX session.

Tag	Field Name	Type	Req	Comments
<*Standard Header*>			Y	MsgType = “5”
58	Text	String	N	Logout reason
<*Standard Trailer*>			Y	

6.3 Resend Request (MsgType = 2)

The message is used to recover an inbound session sequence if a message was missed.

Tag	Field Name	Type	Req	Comments
<*Standard Header*>			Y	MsgType = “2”
7	BeginSeqNo	SeqNum	Y	Sequence number of the first message in range to be resent
16	EndSeqNo	SeqNum	Y	Sequence number of the last message in range to be resent
<*Standard Trailer*>			Y	

6.4 Sequence Reset (MsgType = 4)

The message may be used in two modes:

- **Reset Mode** forces counterparty to adjust inbound message sequence, GapFillFlag = “N” or omitted.
- **Fill Gap Mode** is used during retransmission of messages missed by a client. Administrative messages and rejected business messages are not to be retransmitted. Instead a Sequence Reset message with GapFillFlag = “Y” must be used.

Tag	Field Name	Type	Req	Comments
<*Standard Header*>			Y	MsgType = “4”
123	GapFillFlag	Boolean	N	“N” = sequence reset, the counterparty must adjust inbound sequence number. “Y” = indicates the message is used instead of administrative or business messages which are not to be resent
36	NewSeqNo	SeqNum	Y	New adjusted sequence number
<*Standard Trailer*>			Y	

6.5 Test Request (MsgType = 1)

A Test message is useful for checking sequence numbers or verifying the communication line status. Connection participant received the message is required to reply with Heartbeat message referring to TestReqID of the initial message.

Tag	Field Name	Type	Req	Comments
<*Standard Header*>			Y	MsgType = “1”
112	TestReqID	String	Y	Identifier to be returned in resulting Heartbeat message
<*Standard Trailer*>			Y	

6.6 Heartbeat (MsgType = 0)

Used for replying to the Test request as well as checking the status of communication.

Tag	Field Name	Type	Req	Comments
<Standard Header>			Y	MsgType = “0”
112	TestReqID	String	C	Identifier of a Test Request. Required when the message is the result of a Test Request
<Standard Trailer>			Y	

6.7 Reject (MsgType = 3)

Issued by a party if an incoming FIX message is unsupported or not property formed. Rejected messages must not be resent if a Resend Request is received; instead a SequenceReset with GapFillFlag = “Y” is expected.

Tag	Field Name	Type	Req	Comments
<Standard Header>			Y	MsgType = “3”
45	RefSeqNum	SeqNum	Y	Sequence number of the rejected message
58	Text	String	Y	Error message
373	SessionRejectReason	Int	Y	Reject reason code. See Session Reject Reason Codes (tag 373).
<Standard Trailer>			Y	

7 Order Management Messages

7.1 Overview

The Small Exchange Order Management FIX connection supports the following client-originated messages:

- NewOrderSingle message to submit new single-legged order
- NewOrderMultileg message to submit new order with multiple legs
- OrderCancelRequest message to cancel submitted single or multi-leg order
- OrderCancel/ReplaceRequest message to replace single order
- MultilegOrderCancel/Replace message to replace multi-leg order
- OrderStatusRequest message to receive current order status

The FIX connection sends Execution Report (MsgType = 8) messages to update the connected FIX client with order events:

- New order placed (ExecType = 0) – sent as a reply and referring to the NewOrderSingle or NewOrderMultileg messages to affirm the order is accepted in order book and working. The report is sent for every accepted order even for orders matched immediately.
- Order is cancelled (ExecType = 4) – sent as a reply and referring to OrderCancelRequest message to acknowledge that the cancel is accepted and the original order is no longer working. After receiving the message the original order is assumed to be cancelled, note that no separate individual execution is sent for the cancelled order itself.
- Order is replaced (ExecType = 5) – sent as a reply and referring to OrderCancelRequest or MultilegOrderCancel/Replace messages to affirm the replace is accepted. After receiving the message the original order is assumed to be replaced, no separate individual execution is sent for the replaced order itself.
- Stop order is triggered (ExecType = L) – sent when a stop order is triggered and is placed in the order book
- Order is rejected (ExecType = 8) – sent in case a new or cancel/replace order message is rejected
- Trade (ExecType = F) – sent whenever an order is filled completely or partially
- Order status (ExecType = I) – sent as a reply to an Order Status Request
- Done for Day (ExecType = 3) – sent for all currently open orders after the trading day closes
- Expired (ExecType = C) – sent for the expired Day orders

For rejected cancel requests the Small Exchange API sends the Order Cancel Reject (MsgType = 9) message. Business Reject (MsgType = j) is sent for application-level rejection of any client request which cannot be reported with a well-formed 'rejected' Execution Report.

7.2 Parties Component

This component is used to identify auxiliary information about the financial transaction.

Tag	Field Name	Type	Req	Comments
Repeating group 453	NoPartyIDs	Int	Y	*Number of 'Party' elements in the repeating group*
448	PartyID	String	Y	Identification of the party. Value depends on the PartyRole (452) tag value:

				- Operator ID (for PartyRole = 44). Maximum length is 18 characters. - Subscriber token (for PartyRole = 2101). Maximum length is 15 characters.
447	PartyIDSource	Char	Y	Always 'D' – proprietary / custom code
452	PartyRole	Int	Y	Type or role of the party ID. See PartyRole (tag 452) for the values.

7.3 Instrument Component

The component represents common instrument information in order messages:

Tag	Field Name	Type	Req	Comments
55	Symbol	String	C	Exchange futures or option symbol. Not specified for multi-leg instruments
167	SecurityType	String	Y	FUT = single futures OPT = single option MLEG = multi-leg instrument

7.4 Instrument Leg Component

The following tags are used to identify a leg of a complex instrument in order-related messages:

Tag	Field Name	Type	Req	Comments
600	LegSymbol	String	Y	Instrument symbol of futures or option leg
623	LegRatioQty	Qty	Y	The ratio of this leg. Leg quantity for this leg is OrderQty * LegRatioQty. Max value for option multi-leg is 3.
624	LegSide	Char	Y	Side of the leg. See Order Sides (tag 54 and tag 624).
564	LegPositionEffect	Char	N	Indicates the whether the resulting position after a trade is intended to be opening or closing. If not specified treated as 'D' – default. The Exchange does not validate this tag. See Position Effect (tag 77 and tag 564).
654	LegRefID	String	N	Leg ID specified by a client. Reported by the Exchange back in execution report messages. Up to 8 characters length.

The Exchange requires LegRatioQty values to be normalized:

- OrderQty should be the greatest common factor of individual leg quantities. For example, an order to simultaneously buy 1 InstrumentA and 2 InstrumentB, OrderQty 10 should not be entered as "buy 10 InstrumentA and 20 InstrumentB, Order Qty 1".
- LegRatioQty values should be divided by their greatest common denominator to be accepted. For example, the Exchange will accept 1:2 ratios but won't accept 2:4.

7.5 New Order Single (MsgType = D)

Used to send new single-legged futures or option orders for execution.

Tag	Field Name	Type	Req	Comments
<*Standard Header*>			Y	MsgType = "D"
<*Parties*>			Y	Operator ID (conveyed via PartyRole = 44) is mandatory.
1	Account	String	Y	Identifier of the order's account. Length up to 10 characters.
11	ClOrdID	String	Y	Unique identifier of an order assigned by the client system. Length up to 20 characters. See Order Identifiers.
<*Order Instrument Identity*>			Y	Single-leg futures or option instrument identity
38	OrderQty	Qty	Y	Order quantity
40	OrdType	Char	Y	Order type. See Order Types (tag 40). Market orders are not accepted during opening auction.
44	Price	Price	C	Price per unit of quantity. Required for limit and stop-limit orders
54	Side	Char	Y	Order side. See Order Sides (tag 54 and tag 624).
59	TimeInForce	Char	N	Specifies how long the order remains in effect. See Order Time-in-force (tag 59). Absence of this field indicates Day order.
60	TransactTime	UTC Timestamp	Y	Time this order request was initiated by the trader or trading system
77	PositionEffect	Char	N	The desired order position effect. See Position Effect (tag 77 and tag 564). If not specified treated as 'D' – default. The Exchange does not validate this tag.
99	StopPx	Price	C	Stop price of the order. Required for stop and stop-limit orders.
110	MinQty	Qty	C	Minimum quantity of an order to be executed. Required for Immediate Or Cancel orders. Should be greater than 0.
528	OrderCapacity	Char	Y	Capacity Origin Code. Identifies the counterparty for surveillance/clearing purposes. See OrderCapacity (tag 528).
582	CustOrderCapacity	Int	Y	Customer Type Indicator Code (CTI) as defined by the NFA. Distinguishes for whom and on what type of account the trade is being placed. See CustOrderCapacity (tag 582).
2362	SelfMatchPreventionID	String	N	Self-match prevention token. See Self-Match Prevention. If omitted, default value configured for the connection is used. Both this value and SelfMatchPreventionStrategy (8000) must be defined either in connection-level configuration or in the message.
8000	SelfMatchPrevention Strategy	Char	N	Self-match prevention strategy – see Self-Match Prevention for details. See

				SelfMatchPreventionStrategy (tag 8000) for supported values. If omitted, default value configured for the connection is used. Both this value and SelfMatchPreventionID (2362) must be defined either in connection-level configuration or in the message.
1028	ManualOrderIndicator	Boolean	Y	Indicates if the order was initially received manually (as opposed to electronically)
<Standard Trailer>			Y	

LIMIT futures order example:
BeginString = FIX.4.4
BodyLength = …
MsgType = D
MsgSeqNum = 3
SenderCompID = dx-fix-trade1p
SenderSubID = DX
SendingTime = 20191102-10:15:41.383
TargetCompID = SMALLEX
TargetSubID = TEST
NoPartyIDs = 1
PartyID = trader11
PartyIDSource = D
PartyRole = 44
Account = 658493943
ClOrdID = 314bb362:109f840f9c0
Symbol=SM75F19
SecurityType=FUT
OrderQty = 250
OrdType = 2
Price = 52.51
Side = 1
TimeInForce = 2
TransactTime = 20191102-10:15:40.383
PositionEffect = O
OrderCapacity=A
CustOrderCapacity = 1
SelfMatchPreventionID = TOKEN65
SelfMatchPreventionStrategy = B
ManualOrderIndicator = Y
CheckSum = …

LIMIT option order example:
BeginString = FIX.4.4
BodyLength = …
MsgType = D
MsgSeqNum = 25
SenderCompID = dx-fix-trade1p
SenderSubID = DX
SendingTime = 20200922-09:18:43.011
TargetCompID = SMALLEX
TargetSubID = TEST
NoPartyIDs = 1
PartyID = trader11

PartyIDSource = D
PartyRole = 44
Account = 658493943
ClOrdID = 314bb362:109f840f9c0
Symbol=SM7520Z20201218C45.5
SecurityType=OPT
OrderQty = 2
OrdType = 2
Price = 2.95
Side = 1
TimeInForce = 2
TransactTime = 20200922-09:18:42.302
PositionEffect = O
OrderCapacity=A
CustOrderCapacity = 1
SelfMatchPreventionID = TOKEN65
SelfMatchPreventionStrategy = B
ManualOrderIndicator = Y
CheckSum = …

7.6 New Order Multi-leg (MsgType = AB)

This message is used to send new multi-leg orders for execution.

The Small Exchange order management APIs supports a **long** form of sending multi-leg order. In this form the complex 'symbol' of the multi-leg instrument is not specified and the client sends all the legs of the instrument explicitly. A minimum of 2 legs must be defined, and the API will reject orders with more than 4 legs. Each leg must be defined as Instrument Leg Component.

Supported futures multi-legs (both legs are futures):

- Maximum number of legs is 2
- Maximum leg ratio value is 5 for cross-product orders
- Intra-commodity orders (with a same product) support only calendar leg ratios (buy 1, sell 1)

Supported options multi-legs (at least one leg is option):

- Maximum number of legs is 4
- Maximum leg ratio value is 3
- All legs are for a same product
- Only one leg is allowed to be of a FUT type (for covered call / protected put multi-leg orders)

Side (54) is required for multi-leg orders. Along with Price (44), side defines whether an order is debit (the customer is willing to pay) or credit (the customer is willing to receive money):

- Buy, positive price – debit
- Buy, negative price – credit
- Sell, positive price – credit
- Sell, negative price – debit

In the instrument group the subject of trading is defined – a set of legs defining a spread instrument. Market participants might send the same spread instrument in direct (e.g. Buy A-B) or inverse way (e.g. Buy B-A). These orders will end up in the same CLOB and are normalized to a single internal representation. All executions will be reported to the participants according to the way the order is placed.

Example:
A Buy(A-B) and Buy(B-A) orders are issued. Internally these might be converted to Buy(A-B) and Sell(A-B) orders, however executions will be reported according to initial order structure.

Tag	Field Name	Type	Req	Comments
<*Standard Header*>			Y	MsgType = "AB"
<*Parties*>			Y	Operator ID (conveyed via PartyRole = 44) is mandatory.
1	Account	String	Y	Identifier of the customer as defined by the firm. Maximum length is 10 characters.
11	ClOrdID	String	Y	Unique identifier of the order assigned by the firm. Supports up to 20 characters length. See Order Identifiers.
38	OrderQty	Qty	Y	Order quantity
40	OrdType	Char	Y	Order type. See Order Types (tag 40). Market orders cannot be entered during opening auction.

44	Price	Price	C	Net price of the spread. Required for Limit and Stop-Limit orders.
54	Side	Char	Y	Order side. See Order Sides (tag 54 and tag 624).
59	TimeInForce	Char	N	Specifies how long the order remains in effect. See Order Time-in-force (tag 59). Absence of this field indicates Day order.
60	TransactTime	UTC Timestamp	Y	Time this order request was initiated by trader or trading system
99	StopPx	Price	C	Stop price of the spread. Required for Stop and Stop-Limit orders
110	MinQty	Qty	C	Minimum quantity of the order to be executed. Required for Immediate Or Cancel orders. Should be greater than 0.
528	OrderCapacity	Char	Y	Capacity Origin Code. Identifies the counterparty for surveillance/clearing purposes. See OrderCapacity (tag 528).
582	CustOrderCapacity	Int	Y	Customer Type Indicator Code (CTI) as defined by the NFA. Distinguishes for whom and on what type of account the trade is being placed. See CustOrderCapacity (tag 582).
2362	SelfMatchPreventionID	String	N	Self-match prevention token. See Self-Match Prevention. If omitted, default value configured for the connection is used. Both this value and SelfMatchPreventionStrategy (8000) must be defined either in connection-level configuration or in the message.
8000	SelfMatchPrevention Strategy	Char	N	Self-match prevention strategy – see Self-Match Prevention for details. See SelfMatchPreventionStrategy (tag 8000) for supported values. If omitted, default value configured for the connection is used. Both this value and SelfMatchPreventionID (2362) must be defined either in connection-level configuration or in the message.
1028	ManualOrderIndicator	Boolean	Y	Indicates if the order was initially received manually (as opposed to electronically)
<Order Instrument Identity>			Y	With SecurityType = ”MLEG”
555	NoLegs	NumInGroup	Y	Number of instrument legs. Minimum is 2, maximum is 4.
>	*<Instrument Leg Identity>*		Y	Instrument of the leg
<Standard Trailer>			Y	

LIMIT multi-leg futures order example:
BeginString = FIX.4.4
BodyLength = …
MsgType = AB
MsgSeqNum = 3
SenderCompID = dx-fix-trade1p
SenderSubID = DX
SendingTime = 20191102-10:15:41.383
TargetCompID = SMALLEX
TargetSubID = TEST
NoPartyIDs = 1
PartyID = trader11
PartyIDSource = D
PartyRole = 44
Account = 659848944
ClOrdID = 314bb362:109f840f9c0
OrderQty = 5
OrdType = 2
Price = 52.51
TimeInForce = 2
TransactTime = 20190304-10:15:40.383
OrderCapacity = A
CustOrderCapacity = 1
SelfMatchPreventionID = TOKEN65
SelfMatchPreventionStrategy = B
ManualOrderIndicator = Y
SecurityType = MLEG
NoLegs = 2
LegSymbol = SM75F19
LegRatioQty = 1
LegSide = 1
LegSymbol = SM75G19
LegRatioQty = 1
LegSide = 2
CheckSum = …

LIMIT multi-leg option vertical order example:
BeginString = FIX.4.4
BodyLength = …
MsgType = AB
MsgSeqNum = 28
SenderCompID = dx-fix-trade1p
SenderSubID = DX
SendingTime = 20200922-10:15:41.383
TargetCompID = SMALLEX
TargetSubID = TEST
NoPartyIDs = 1
PartyID = trader11
PartyIDSource = D
PartyRole = 44
Account = 658923544
ClOrdID = 314bb362:109f840f9c0
OrderQty = 5
OrdType = 2
Price = 1.51

TimeInForce = 2
TransactTime = 20200922-10:15:40.383
OrderCapacity = A
CustOrderCapacity = 1
SelfMatchPreventionID = TOKEN65
SelfMatchPreventionStrategy = B
ManualOrderIndicator = Y
SecurityType = MLEG
NoLegs = 2
LegSymbol = SM7520Z20201218C51.5
LegRatioQty = 1
LegSide = 1
LegSymbol = SM7520Z20201218C52.5
LegRatioQty = 1
LegSide = 2
CheckSum = …

7.7 Order Cancel Request (MsgType = F)

A firm may use this message to cancel the remaining quantity of any working single or multi-leg order.

Tag	Field Name	Type	Req	Comments
<*Standard Header*>			Y	MsgType = "F"
<*Parties*>			Y	Operator ID (conveyed via PartyRole = 44) is mandatory.
1	Account	String	Y	Identifier of original order's account. Supports up to 10 characters length.
11	ClOrdID	String	Y	Cancel order unique identifier assigned by the firm. Maximum length 20 characters. See Order Identifiers.
37	OrderID	String	N	The Exchange identifier of the original order. This value is supplied by the Exchange on order acknowledgement. This value is optional but recommended for better performance. Maximum length is 20 characters.
38	OrderQty	Qty	N	Required field by FIX protocol specification. Can be optionally specified although it is not processed by the exchange.
41	OrigClOrdID	String	Y	Order ClOrdID for identifying the previous order in cancel and cancel/replace requests. Supports up to 20 characters length.
54	Side	Char	Y	Side of order to be cancelled. This value is ignored.
1028	ManualOrder Indicator	Boolean	Y	Indicates if the cancel order was initially received manually (as opposed to electronically)
<Order Instrument Identity>			Y	Symbol and SecurityType are required for cancelling single orders. For multi-leg order cancels only SecurityType="MLEG" is required.
60	TransactTime	UTC Timestamp	Y	Time this cancel request was initiated by trader or trading system

<Standard Trailer>		Y	

Cancel order request example:
BeginString = FIX.4.4
BodyLength = …
MsgType = F
MsgSeqNum = 6
SenderCompID = dx-fix-trade1p
SenderSubID = DX
SendingTime = 20191102-10:15:41.383
TargetCompID = SMALLEX
TargetSubID = TEST
NoPartyIDs = 1
PartyID = trader11
PartyIDSource = D
PartyRole = 44
Account = 65
ClOrdID = 65780bfc:901n328943d
OrderID = 5038
OrigClOrdID = 71fa0cda:109f847292d
Side = 1
ManualOrderIndicator = Y
Symbol = SM75G19
SecurityType = FUT
TransactTime = 20191102-10:15:40.725
CheckSum = …

7.8 Order Cancel/Replace Request (MsgType = G)

A firm may use this message to replace an order. The following restrictions exist:

- Triggered Stop and Stop-Limit orders cannot be replaced, only canceled
- It is possible to change the order type, but only from Limit to Market
- Time in Force can be changed, but only GTC → DAY and DAY→ GTC replaces are supported
- If a field cannot be changed, the replacing order gets rejected.

Tag	Field Name	Type	Req	Comments
<Standard Header>			Y	MsgType = “G”
<Parties>			Y	Operator ID (conveyed via PartyRole = 44) is mandatory.
1	Account	String	Y	Identifier of original order’s account. Supports up to 10 characters length. Account cannot be changed.
11	ClOrdID	String	Y	Unique identifier of the replacing order assigned by the client system. Maximum length 20 characters. See Order Identifiers.
<Order Instrument Identity>			Y	Instrument cannot be changed.
38	OrderQty	Qty	Y	Quantity of the replacing order
40	OrdType	Char	Y	Order type of the replacing order. See Order Types (tag 40). Order type can only be changed from Limit to Market.
41	OrigClOrdID	String	Y	ClOrdID of the original order to be replaced.

				Maximum length 20 characters. See Order Identifiers.
44	Price	Price	C	Price per unit of quantity of the replacing order. Required for Limit and Stop-Limit orders
54	Side	Char	Y	Side of the order to be replaced. See Order Sides (tag 54 and tag 624). Order side cannot be changed.
59	TimeInForce	Char	Y	Specifies how long the order remains in effect. See Order Time-in-force (tag 59). Only GTC → DAY and DAY→ GTC replaces are supported.
60	TransactTime	UTC Timestamp	Y	Time this replace request was initiated by trader or trading system
77	PositionEffect	Char	N	The desired order position effect. See Position Effect (tag 77 and tag 564). If not specified treated as 'D' – default. The Exchange does not validate this tag.
99	StopPx	Price	C	Stop price of the order. Required for Stop and Stop-Limit orders
110	MinQty	Qty	C	Minimum quantity of an order to be executed. Required for Immediate Or Cancel orders. Should be greater than 0.
528	OrderCapacity	Char	Y	Capacity Origin Code. Identifies the counterparty for surveillance/clearing purposes. See OrderCapacity (tag 528).
582	CustOrderCapacity	Int	Y	Customer Type Indicator Code (CTI) as defined by the NFA. Distinguishes for whom and on what type of account the trade is being placed. See CustOrderCapacity (tag 582).
2362	SelfMatchPrevention ID	String	N	New self-match prevention token. See Self-Match Prevention. If omitted, default value configured for the connection is used. Both this value and SelfMatchPreventionStrategy (8000) must be defined either in connection-level configuration or in the message.
8000	SelfMatchPrevention Strategy	Char	N	Updated self-match prevention strategy – see Self-Match Prevention for details. See SelfMatchPreventionStrategy (tag 8000) for supported values. If omitted, default value configured for the connection is used. Both this value and SelfMatchPreventionID (2362) must be defined either in connection-level configuration or in the message.
1028	ManualOrder Indicator	Boolean	Y	Indicates if the order was initially received manually (as opposed to electronically)
<Standard Trailer>			Y	

Cancel/Replace request of a LIMIT order example:
BeginString = FIX.4.4
BodyLength = …
MsgType = G
MsgSeqNum = 6
SenderCompID = dx-fix-trade1p
SenderSubID = DX
SendingTime = 20191102-10:15:41.383
TargetCompID = SMALLEX
TargetSubID = TEST
NoPartyIDs = 1
PartyID = trader11
PartyIDSource = D
PartyRole = 44
Account = 65
ClOrdID = 71fa0cda:109f847292d:-7ffe
SecurityType = FUT
Symbol=SM75F19
OrderQty = 150
OrdType = 2
OrigClOrdID = 71fa0cda:109f847292d:-7ffd
Price = 49.6
Side = 1
TimeInForce = 2
TransactTime = 20191102-10:15:40.725
PositionEffect = O
OrderCapacity = A
CustOrderCapacity = 1
SelfMatchPreventionID = TOKEN65
SelfMatchPreventionStrategy = B
ManualOrderIndicator = Y
CheckSum = …

7.9 Multi-leg Order Cancel/Replace Request (MsgType = AC)

A firm may use this message to replace a multi-leg order. The following restrictions exist:

- Triggered Stop and Stop-Limit orders cannot be replaced, only canceled
- It is possible to change the order type, but only from Limit to Market
- Time in Force can be changed, but only GTC → DAY and DAY→ GTC replaces are supported
- If a field cannot be changed, the replacing order gets rejected.

Price of the multi-leg order is required for Limit and Stop-Limit orders. Sign of the price value identifies whether the order is **debit** or **credit**:

- Debit (positive) – the customer is willing to pay
- Credit (negative) – the customer is willing to receive money
- 0 (Zero) – even orders

Tag	Field Name	Type	Req	Comments
<Standard Header>			Y	MsgType = “AC”
<*Parties*>			Y	Operator ID (conveyed via PartyRole = 44) is mandatory.
1	Account	String	Y	Identifier of original order’s account. Supports up to 10 characters length. Account cannot be changed.

11	ClOrdID	String	Y	Unique identifier of the replacing order assigned by the client system. Maximum length 20 characters. See Order Identifiers.
38	OrderQty	Qty	Y	Quantity of the replacing order
40	OrdType	Char	Y	Order type of the replacing order. See Order Types (tag 40). Order type can only be changed from Limit to Market.
41	OrigClOrdID	String	Y	ClOrdID of the original order to be replaced. Maximum length 20 characters. See Order Identifiers.
44	Price	Price	C	Net price of the spread. Required for Limit and Stop-Limit orders.
54	Side	Char	Y	Side of the order to be replaced. See Order Sides (tag 54 and tag 624). Order side cannot be changed.
59	TimeInForce	Char	Y	Specifies how long the order remains in effect. See Order Time-in-force (tag 59). Only GTC → DAY and DAY→ GTC replaces are supported.
60	TransactTime	UTC Timestamp	Y	Time this order request was initiated by trader or trading system
99	StopPx	Price	C	Stop price of the order. Required for Stop and Stop-Limit orders.
110	MinQty	Qty	C	Minimum quantity of an order to be executed. Required for Immediate Or Cancel orders. Should be greater than 0.
528	OrderCapacity	Char	Y	Capacity Origin Code. Identifies the counterparty for surveillance/clearing purposes. See OrderCapacity (tag 528).
582	CustOrderCapacity	Int	Y	Customer Type Indicator Code (CTI) as defined by the NFA. Distinguishes for whom and on what type of account the trade is being placed. See CustOrderCapacity (tag 582).
2362	SelfMatchPrevention ID	String	N	New self-match prevention token. See Self-Match Prevention. If omitted, default value configured for the connection is used. Both this value and SelfMatchPreventionStrategy (8000) must be defined either in connection-level configuration or in the message.
8000	SelfMatchPrevention Strategy	Char	N	Updated self-match prevention strategy – see Self-Match Prevention for details. See SelfMatchPreventionStrategy (tag 8000) for supported values. If omitted, default value configured for the connection is used. Both this value and SelfMatchPreventionID (2362)

				must be defined either in connection-level configuration or in the message.
1028	ManualOrder Indicator	Boolean	Y	Indicates if the order was initially received manually (as opposed to electronically)
<Order Instrument Identity>			Y	With SecurityType = ”MLEG”. Instrument cannot be changed.
555	NoLegs	NumInGroup	Y	Number of instrument legs
>	*<Instrument Leg Identity>*		Y	Instrument of the leg
<Standard Trailer>			Y	

Cancel/Replace request of a multi-leg LIMIT futures order example:
BeginString = FIX.4.4
BodyLength = …
MsgType = AC
MsgSeqNum = 6
SenderCompID = dx-fix-trade1p
SenderSubID = DX
SendingTime = 20191102-10:15:41.383
TargetCompID = SMALLEX
TargetSubID = TEST
NoPartyIDs = 1
PartyID = trader11
PartyIDSource = D
PartyRole = 44
Account = 65
ClOrdID = 71fa0cda:109f847292d:-7ffe
OrderQty = 150
OrdType = 2
OrigClOrdID = 71fa0cda:109f847292d:-7ffd
Price = 49.6
Side = 1
TimeInForce = 2
TransactTime = 20191102-10:15:40.725
OrderCapacity = A
CustOrderCapacity = 1
SelfMatchPreventionID = TOKEN65
SelfMatchPreventionStrategy = B
ManualOrderIndicator = Y
SecurityType = MLEG
NoLegs = 2
LegSymbol = SM75F19
LegRatioQty = 1
LegSide = 1
LegSymbol = SM75G19
LegRatioQty = 1
LegSide = 2
CheckSum = …

7.10 Order Status Request (MsgType = H)

The message used to request the current status of an existing order reported by the Small Exchange system as Execution Report message with ExecType=I.

Tag	Field Name	Type	Req	Comments
<Standard Header>			Y	MsgType = “H”
11	ClOrdID	String	Y	Order unique identifier assigned by client system. Maximum length 20 characters. See Order Identifiers.
37	OrderID	String	Y	The Exchange identifier of the original order. This value is supplied by the Exchange on order acknowledgement. Unlike in the general FIX protocol specification the field is mandatory in the Small Exchange FIX API. Maximum length is 20 characters.
<Order Instrument Identity>			Y	
54	Side	Char	Y	Side of the order. See Order Sides (tag 54 and tag 624).
790	OrdStatusRequestID	String	Y	Unique identifier of an Order Status Request message. Maximum length 20 characters.
<Standard Trailer>			Y	

Order Status Request example:
BeginString = FIX.4.4
BodyLength = …
MsgType = H
SenderCompID = dx-fix-trade1p
SenderSubID = DX
SendingTime = 20191102-10:15:41.383
TargetCompID = SMALLEX
TargetSubID = TEST
MsgSeqNum = 2
ClOrdID = 4c2a1e5e:10c2002c007
OrderID = 5038
SecurityType=FUT
Symbol=SM75F19
Side = 1
OrdStatusRequestID = 77f7de35
CheckSum = …

7.11 Execution Report: Accepted Order (MsgType = 8, ExecType = 0 or 4 or 5)

Reports with execution type New (0), Cancelled (4), Replaced (5) represent an acknowledgement of successful acceptance of a New, Cancel, Cancel/Replace order or Buy/Sell side of accepted Mass Quote Entry.

Tag	Field Name	Type	Req	Comments
<*Standard Header*>			Y	MsgType = "8"
<*Parties*>			Y	Operator ID (PartyRole = 44)
1	Account	String	Y	Unique identifier of an account in the system (as specified in the order).
6	AvgPx	Price	Y	Always "0"
11	ClOrdID	String	Y	Accepted order identifier assigned by the client system (as defined in the order). See Order Identifiers.
14	CumQty	Qty	Y	Total traded quantity for the order chain (current order and all replaced orders in the chain)
17	ExecID	String	Y	Unique identifier of the execution as assigned by the Exchange. Maximum length is 20 characters.
37	OrderID	String	Y	Unique identifier of an order chain in the Small Exchange system. Maximum length is 20 characters.
38	OrderQty	Qty	Y	Quantity ordered.
39	OrdStatus	Char	Y	Identifies the current status of an order. See Order Status (tag 39).
40	OrdType	Char	Y	Order type as specified in the order. See Order Types (tag 40).
41	OrigClOrdID	String	C	ClOrdID of replaced/cancelled order. Reported for cancel and cancel/replace orders acknowledgements. See Order Identifiers.
44	Price	Price	C	Price of the Limit or Stop-Limit order. Reported if specified for the order.
54	Side	Char	Y	Order side. See Order Sides (tag 54 and tag 624).
59	TimeInForce	Char	Y	Specifies how long the order remains in effect. See Order Time-in-force (tag 59).
60	TransactTime	UTC Timestamp	Y	Time of execution, expressed in UTC
75	TradeDate	Local MktDate	Y	Business date of the execution.
99	StopPx	Price	C	Order stop price. Reported if specified for the order.
110	MinQty	Qty	C	Minimum quantity of an order to be executed. Reported if specified for the order. Required for IOC orders. Should be greater than 0.
150	ExecType	Char	Y	"0" = new order acknowledgement "4" = cancel order acknowledgement

				"5" = cancel/replace order acknowledgement
151	LeavesQty	LeavesQty	Y	Quantity available for further execution, working quantity
377	SolicitedFlag	Boolean	N	Indicates whether or not the report was solicited. This flag is set when an order has been cancelled by the risk management procedures on the Exchange (kill switch activation etc.).
528	OrderCapacity	Char	Y	Capacity Origin Code copied from the order. Identifies the counterparty for surveillance/clearing purposes. See OrderCapacity (tag 528).
582	CustOrderCapacity	Int	Y	Customer Type Indicator Code (CTI) as defined by the NFA (copied from the order). Distinguishes for whom and on what type of account the trade is being placed. See CustOrderCapacity (tag 582).
2362	SelfMatchPreventionID	String	N	Self-match prevention token as defined in the order. See Self-Match Prevention.
8000	SelfMatchPrevention Strategy	Char	N	Self-match prevention strategy – see Self-Match Prevention for details. See SelfMatchPreventionStrategy (tag 8000) for supported values.
1028	ManualOrderIndicator	Boolean	Y	Indicates if the order was initially received manually (as opposed to electronically)
1166	QuoteMsgID	String	N	Refers to QuoteID<117> of a mass quote message originally placed the order
<Order Instrument Identity>			Y	
555	NoLegs	NumInGroup	C	Number of instrument legs. Only for multi leg orders
>	*<Instrument Leg Identity>*		C	Instrument of the leg
<Standard Trailer>			Y	

7.12 Execution Report: Order Status (MsgType = 8, ExecType = I)

Reported as a reply to order status request.

Tag	Field Name	Type	Req	Comments
<Standard Header>			Y	MsgType = "8"
<Parties>			Y	Operator ID (PartyRole = 44)
1	Account	String	Y	Unique identifier of an account in the system (as specified in the order).
6	AvgPx	Price	Y	Always "0"
11	ClOrdID	String	Y	Accepted order identifier assigned by the client system (as defined in the order). See Order Identifiers.

14	CumQty	Qty	Y	Total traded quantity for the order chain (current order and all replaced orders in the chain)
17	ExecID	String	Y	Unique identifier of the execution as assigned by the Exchange. Maximum length is 20 characters.
37	OrderID	String	Y	Unique identifier of an order chain in the Small Exchange system. Maximum length is 20 characters.
38	OrderQty	Qty	Y	Quantity ordered.
39	OrdStatus	Char	Y	Identifies the current status of an order. See Order Status (tag 39).
40	OrdType	Char	Y	Order type as specified in the order. See Order Types (tag 40).
44	Price	Price	C	Price of the Limit or Stop-Limit order. Reported if specified for the order.
54	Side	Char	Y	Order side. See Order Sides (tag 54 and tag 624).
59	TimeInForce	Char	Y	Specifies how long the order remains in effect. See Order Time-in-force (tag 59).
60	TransactTime	UTC Timestamp	Y	Time of execution, expressed in UTC
75	TradeDate	Local MktDate	Y	Business date of the execution.
99	StopPx	Price	C	Order stop price. Reported if specified for the order.
110	MinQty	Qty	C	Minimum quantity of an order to be executed. Reported if specified for the order. Required for IOC orders. Should be greater than 0.
150	ExecType	Char	Y	“I” = order status
151	LeavesQty	LeavesQty	Y	Working quantity of the order
377	SolicitedFlag	Boolean	N	Indicates whether or not the report was solicited. Y = was solicited N = was not solicited This flag is set when an order has been cancelled by the risk management procedures on the Exchange (kill switch activation etc.).
790	OrdStatusRequestID	String	Y	OrdStatusRequestID value copied from the order status request
528	OrderCapacity	Char	Y	Capacity Origin Code copied from the order. Identifies the counterparty for surveillance/clearing purposes. See OrderCapacity (tag 528).
582	CustOrderCapacity	Int	Y	Customer Type Indicator Code (CTI) as defined by the NFA (copied from the order). Distinguishes for whom and on what type of account the trade is being placed. See CustOrderCapacity (tag 582).

2362	SelfMatchPreventionID	String	N	Self-match prevention token as defined in the order. See Self-Match Prevention. This can be specified in connection-level configuration.
8000	SelfMatchPrevention Strategy	C	N	Self-match prevention strategy – see Self-Match Prevention for details. See SelfMatchPreventionStrategy (tag 8000) for supported values. This can be specified in connection-level configuration.
1028	ManualOrderIndicator	Boolean	Y	Indicates if the order was initially received manually (as opposed to electronically)
<Order Instrument Identity>			Y	
555	NoLegs	NumInGroup	C	Number of instrument legs. Only for multi leg orders
>	*<Instrument Leg Identity>*		C	Instrument of the leg
<Standard Trailer>			Y	

7.13 Execution Report: Triggered Stop (MsgType = 8, ExecType = L)

Sent by the Small Exchange to report a triggered stop order event. Note that triggered stop orders do not change their type and are not converted to limit or market orders.

Tag	Field Name	Type	Req	Comments
<*Standard Header*>			Y	MsgType = "8"
<*Parties*>			Y	Operator ID (PartyRole = 44)
1	Account	String	Y	Unique identifier of an account in the system (as specified in the order).
6	AvgPx	Price	Y	Always "0"
11	ClOrdID	String	Y	Accepted order identifier assigned by the client system (as defined in the order). See Order Identifiers.
14	CumQty	Qty	Y	Total traded quantity for the order chain (current order and all replaced orders in the chain)
17	ExecID	String	Y	Unique identifier of the execution as assigned by the Exchange. Maximum length is 20 characters.
37	OrderID	String	Y	Unique identifier of an order chain in the Small Exchange system. Maximum length is 20 characters.
38	OrderQty	Qty	Y	Order quantity
39	OrdStatus	Char	Y	"0" = New order
40	OrdType	Char	Y	Order type as specified in the order. See Order Types (tag 40).
44	Price	Price	C	Price of the Limit or Stop-Limit order. Reported if specified for the order.
54	Side	Char	Y	Order side. See Order Sides (tag 54 and tag 624).
59	TimeInForce	Char	Y	Specifies how long the order remains in effect. See Order Time-in-force (tag 59).
60	TransactTime	UTC Timestamp	Y	Time of execution, expressed in UTC
75	TradeDate	Local MktDate	Y	Business date of the execution.
99	StopPx	Price	Y	Order stop price.
150	ExecType	Char	Y	"L" = triggered by the system
151	LeavesQty	LeavesQty	Y	Quantity available for further execution, working quantity
528	OrderCapacity	Char	Y	Capacity Origin Code copied from the order. Identifies the counterparty for surveillance/clearing purposes. See OrderCapacity (tag 528).
582	CustOrderCapacity	Int	Y	Customer Type Indicator Code (CTI) as defined by the NFA (copied from the order). Distinguishes for whom and on what type of account the trade is being placed. See CustOrderCapacity (tag 582).
2362	SelfMatchPreventionID	String	N	Self-match prevention token as defined in the order. See Self-Match

				Prevention. This can be specified in connection-level configuration.
8000	SelfMatchPrevention Strategy	Char	N	Self-match prevention strategy – see Self-Match Prevention for details. See SelfMatchPreventionStrategy (tag 8000) for supported values. This can be specified in connection-level configuration.
1028	ManualOrderIndicator	Boolean	Y	Indicates if the order was initially received manually (as opposed to electronically)
<Order Instrument Identity>			Y	
555	NoLegs	NumInGroup	C	Number of instrument legs. Only for multi leg orders
>	*<Instrument Leg Identity>*		C	Instrument of the leg
<Standard Trailer>			Y	

7.14 Execution Report: Trade (MsgType = 8, ExecType = F)

Sent by the Small Exchange system to report order or mass quote entry fill.
Note that multi-leg order fill reports are sent individually for each leg of futures or option order.

Tag	Field Name	Type	Req	Comments
<Standard Header>			Y	MsgType = “8”
<Parties>			Y	Operator ID (PartyRole = 44)
1	Account	String	Y	Unique identifier of an account in the system (as specified in the order).
6	AvgPx	Price	Y	Always “0”
11	ClOrdID	String	Y	Accepted order identifier assigned by the client system (as defined in the order). See Order Identifiers.
14	CumQty	Qty	Y	Total traded quantity for the order chain (current order and all replaced orders in the chain)
17	ExecID	String	Y	Unique identifier of the execution as assigned by the Exchange. Maximum length is 20 characters.
<Order Instrument Identity>			Y	Instrument traded
31	LastPx	Price	Y	Price of the fill
32	LastQty	Qty	Y	Quantity bought or sold in this fill
37	OrderID	String	Y	Unique identifier of an order chain in the Small Exchange system. Maximum length is 20 characters.
38	OrderQty	Qty	Y	Order quantity
39	OrdStatus	Char	Y	Order status. See Order Status (tag 39).
40	OrdType	Char	Y	Order type as specified in the order. See Order Types (tag 40).
44	Price	Price	C	Price of the Limit or Stop-Limit order. Reported if specified for the order.
624	LegRefID	String	C	Unique identifier for a specific leg (included only for spread leg fill reports). References the value from the order request (see Instrument Leg Component).

54	Side	Char	Y	Order side. See Order Sides (tag 54 and tag 624).
59	TimeInForce	Char	Y	Specifies how long the order remains in effect. See Order Time-in-force (tag 59).
60	TransactTime	UTC Timestamp	Y	Time of execution, expressed in UTC
75	TradeDate	Local MktDate	Y	Business date of the trade
99	StopPx	Price	C	Order stop price. Reported if specified for the order.
110	MinQty	Qty	C	Minimum quantity of an order to be executed. Reported if specified for the order.
150	ExecType	Char	Y	“F” = trade
151	LeavesQty	LeavesQty	Y	Remaining working quantity of the order
442	MultiLegReportingType	Char	C	Reported for multi-leg order fills. Always “2” - individual leg of a multi-leg security
528	OrderCapacity	Char	Y	Capacity Origin Code copied from the order. Identifies the counterparty for surveillance/clearing purposes. See OrderCapacity (tag 528).
582	CustOrderCapacity	Int	Y	Customer Type Indicator Code (CTI) as defined by the NFA (copied from the order). Distinguishes for whom and on what type of account the trade is being placed. See CustOrderCapacity (tag 582).
880	TrdMatchID	String	Y	Unique identifier of a trade in the Small Exchange system. Same for BUY and SELL sides report
2362	SelfMatchPreventionID	String	N	Self-match prevention token as defined in the order. See Self-Match Prevention.
8000	SelfMatchPrevention Strategy	Char	N	Self-match prevention strategy – see Self-Match Prevention for details. See SelfMatchPreventionStrategy (tag 8000) for supported values.
1028	ManualOrderIndicator	Boolean	Y	Indicates if the order was initially received manually (as opposed to electronically)
1057	AggressorIndicator	Boolean	Y	Indicates if an order was aggressor or not. Y - aggressor order taking liquidity, N - passive/resting order. Note that all orders matched during an opening auction are reported as passive
1166	QuoteMsgID	String	N	Refers to QuoteID<117> of a mass quote message originally placed the order
<Standard Trailer>			Y	

Trade execution report example: multi-leg option order fill

This example assumes a multi-leg order with the following parameters has been issued:
NoLegs = 2
LegSymbol = SM752020Z20201218C51.5
LegRefID = 16777225
LegPositionEffect = C
LegRatioQty = 1
LegSide = 1
LegSymbol = SM7520Z20201218C52.5
LegRefID = 16777226
LegPositionEffect = C
LegRatioQty = 1
LegSide = 2

First leg fill:

BeginString = FIX.4.4
BodyLength = …
MsgType = 8
SenderCompID = SMALLEX
SenderSubID = TEST
SendingTime = 20201022-10:15:41.383
TargetCompID = dx-fix-trade1p
TargetSubID = DX
MsgSeqNum = 2
NoPartyIDs = 1
PartyID = trader11
PartyIDSource = D
PartyRole = 44
OrderID = 72057594037977283
ClOrdID = 4c2a1e5e:10c2002c007
ExecID = 72057594037981287
ExecType = F
OrdStatus = 2
Account = 432434224
Symbol = SM7520Z20201218C51.5
SecurityType=OPT
LegRefID = 16777225
Side = 1
OrderQty = 1
OrdType = 2
Price = 0
TimeInForce = 0
LastQty = 2
LastPx = 15
LeavesQty = 0
CumQty = 0
AvgPx = 0
TradeDate = 20201022
TransactTime = 20201022-12:38:57.940
MultiLegReportingType = 2
CustOrderCapacity = 1
OrderCapacity = A
ManualOrderIndicator = Y

AggressorIndicator = Y
TrdMatchID = 78654
CheckSum =

Second leg fill:

BeginString = FIX.4.4
BodyLength = …
MsgType = 8
SenderCompID = SMALLEX
SenderSubID = TEST
SendingTime = 20201022-10:15:41.383
TargetCompID = dx-fix-trade1p
TargetSubID = DX
MsgSeqNum = 2
NoPartyIDs = 1
PartyID = trader11
PartyIDSource = D
PartyRole = 44
OrderID = 72057594037977283
ClOrdID = 4c2a1e5e:10c2002c007
ExecID = 72057594037981289
ExecType = F
OrdStatus = 2
Account = fixmleg
Symbol = SM7520Z20201218C52.5
SecurityType=OPT
LegRefID = 16777226
Side = 2
OrderQty = 1
OrdType = 2
Price = 0
TimeInForce = 0
LastQty = 2
LastPx = 14
LeavesQty = 0
CumQty = 0
AvgPx = 0
TradeDate = 20201022
TransactTime = 20201022-12:38:57.940
MultiLegReportingType = 2
CustOrderCapacity = 1
OrderCapacity = A
ManualOrderIndicator = Y
AggressorIndicator = Y
TrdMatchID = 78654
CheckSum =

7.15 Execution Report: Rejected New Order (MsgType = 8, ExecType = 8)

Sent as a reject for new single or multi-leg order request or as a reject to mass quote entry buy/sell side.

Tag	Field Name	Type	Req	Comments
<Standard Header>			Y	MsgType = “8”
<Parties>			Y	Operator ID (PartyRole = 44)
1	Account	String	Y	Unique identifier of an account in the system (as specified in the order).
6	AvgPx	Price	Y	Always “0”
11	ClOrdID	String	Y	Accepted order identifier assigned by the client system (as defined in the order). See Order Identifiers.
17	ExecID	String	Y	Unique identifier of the execution as assigned by the Exchange. Maximum length is 20 characters.
37	OrderID	String	Y	Unique identifier of the execution as assigned by the Exchange. Maximum length is 20 characters.
39	OrdStatus	Char	Y	“8” - rejected
40	OrdType	Char	N	Order type as specified in the order. See Order Types (tag 40).
54	Side	Char	Y	Order side. See Order Sides (tag 54 and tag 624).
58	Text	String	N	Reject reason text
60	TransactTime	UTC Timestamp	Y	Timestamp of the execution report, expressed in UTC
75	TradeDate	Local MktDate	Y	Business date of the execution
103	OrdRejReason	Int	Y	Reject reason code. See Order Reject Reason (tag 103).
110	MinQty	Qty	C	Minimum quantity of an order to be executed. Reported if specified for the order; sent only for IOC orders.
150	ExecType	ExecType	Y	“8” - rejected
151	LeavesQty	LeavesQty	Y	Always “0”
377	SolicitedFlag	Boolean	N	Indicates whether or not the order was solicited, always Y for the rejected order
528	OrderCapacity	Char	Y	Capacity Origin Code copied from the order. Identifies the counterparty for surveillance/clearing purposes. See OrderCapacity (tag 528).
582	CustOrderCapacity	Int	Y	Customer Type Indicator Code (CTI) as defined by the NFA (copied from the order). Distinguishes for whom and on what type of account the trade is being placed. See CustOrderCapacity (tag 582).
2362	SelfMatchPreventionID	String	N	Self-match prevention token as defined in the order. See Self-Match Prevention.

8000	SelfMatchPrevention Strategy	Char	N	Self-match prevention strategy – see Self-Match Prevention for details. See SelfMatchPreventionStrategy (tag 8000) for supported values.
1028	ManualOrderIndicator	Boolean	Y	Indicates if the order was initially received manually (as opposed to electronically)
1166	QuoteMsgID	String	N	Refers to QuoteID<117> of a mass quote message
<Order Instrument Identity>			Y	
555	NoLegs	NumInGroup	C	Number of instrument legs. Only for multi-leg orders
>	*<Instrument Leg Identity>*		C	Instrument of the leg
<Standard Trailer>			Y	

7.16 Execution Report: Done for Day (MsgType = 8, ExecType = 3)

Reports with execution type 'Done for Day' (4) are sent by the Exchange to the firms that need the status of their open orders after the trading session is closed.

Tag	Field Name	Type	Req	Comments
<Standard Header>			Y	MsgType = "8"
<Parties>			Y	Operator ID (PartyRole = 44)
1	Account	String	Y	Unique identifier of an account in the system (as specified in the order).
6	AvgPx	Price	Y	Always "0"
11	ClOrdID	String	Y	The order identifier assigned by the client system (as defined in the order). See Order Identifiers.
14	CumQty	Qty	Y	Total traded quantity for the order chain (current order and all replaced orders in the chain)
17	ExecID	String	Y	Unique identifier of the execution as assigned by the Exchange. Maximum length is 20 characters.
37	OrderID	String	Y	Unique identifier of an order chain in the Small Exchange system. Maximum length is 20 characters.
38	OrderQty	Qty	Y	Quantity ordered.
39	OrdStatus	Char	Y	Identifies the current status of an order. Always "3" = Done for Day. See Order Status (tag 39).
40	OrdType	Char	Y	Order type as specified in the order. See Order Types (tag 40).
44	Price	Price	C	Price of the Limit or Stop-Limit order. Reported if specified for the order.
54	Side	Char	Y	Order side. See Order Sides (tag 54 and tag 624).
59	TimeInForce	Char	Y	Specifies how long the order remains in effect. See Order Time-in-force (tag 59).
60	TransactTime	UTC Timestamp	Y	Time of execution, expressed in UTC
75	TradeDate	Local	Y	Business date of the execution.

		MktDate		
99	StopPx	Price	C	Order stop price. Reported if specified for the order.
110	MinQty	Qty	C	Minimum quantity of an order to be executed. Reported if specified for the order. Required for IOC orders.
150	ExecType	Char	Y	“3” = Done for Day
151	LeavesQty	LeavesQty	Y	Quantity available for further execution, working quantity
528	OrderCapacity	Char	Y	Capacity Origin Code copied from the order. Identifies the counterparty for surveillance/clearing purposes. See OrderCapacity (tag 528).
582	CustOrderCapacity	Int	Y	Customer Type Indicator Code (CTI) as defined by the NFA (copied from the order). Distinguishes for whom and on what type of account the trade is being placed. See CustOrderCapacity (tag 582).
2362	SelfMatchPreventionID	String	N	Self-match prevention token as defined in the order. See Self-Match Prevention.
8000	SelfMatchPrevention Strategy	Char	N	Self-match prevention strategy – see Self-Match Prevention for details. See SelfMatchPreventionStrategy (tag 8000) for supported values.
1028	ManualOrderIndicator	Boolean	Y	Indicates if the order was initially received manually (as opposed to electronically)
<Order Instrument Identity>			Y	
555	NoLegs	NumInGroup	C	Number of instrument legs. Only for multi leg orders
>	*<Instrument Leg Identity>*		C	Instrument of the leg
<Standard Trailer>			Y	

7.17 Execution Report: Expired (MsgType = 8, ExecType = C)

Reports with execution type Expired (C) are sent by the Exchange to the firms that need to track the expiration of their time-limited orders such as DAY or GTC orders on instrument expiration day.

Tag	Field Name	Type	Req	Comments
<Standard Header>			Y	MsgType = “8”
<Parties>			Y	Operator ID (PartyRole = 44)
1	Account	String	Y	Unique identifier of an account in the system (as specified in the order).
6	AvgPx	Price	Y	Always “0”
11	ClOrdID	String	Y	The order identifier assigned by the client system (as defined in the order). See Order Identifiers.
14	CumQty	Qty	Y	Total traded quantity for the order chain (current order and all replaced orders in the chain)

17	ExecID	String	Y	Unique identifier of the execution as assigned by the Exchange. Maximum length is 20 characters.
37	OrderID	String	Y	Unique identifier of an order chain in the Small Exchange system. Maximum length is 20 characters.
38	OrderQty	Qty	Y	Quantity ordered.
39	OrdStatus	Char	Y	Identifies the current status of an order. Always C = Expired. See Order Status (tag 39).
40	OrdType	Char	Y	Order type as specified in the order. See Order Types (tag 40).
44	Price	Price	C	Price of the Limit or Stop-Limit order. Reported if specified for the order.
54	Side	Char	Y	Order side. See Order Sides (tag 54 and tag 624).
59	TimeInForce	Char	Y	Specifies how long the order remains in effect. See Order Time-in-force (tag 59).
60	TransactTime	UTC Timestamp	Y	Time of execution, expressed in UTC
75	TradeDate	Local MktDate	Y	Business date of the execution.
99	StopPx	Price	C	Order stop price. Reported if specified for the order.
110	MinQty	Qty	C	Minimum quantity of an order to be executed. Reported if specified for the order. Required for IOC orders.
150	ExecType	ExecType	Y	“C” – Expired.
151	LeavesQty	LeavesQty	Y	Quantity available for further execution, working quantity. Always “0”.
528	OrderCapacity	Char	Y	Capacity Origin Code copied from the order. Identifies the counterparty for surveillance/clearing purposes. See OrderCapacity (tag 528).
582	CustOrderCapacity	Int	Y	Capacity Origin Code copied from the order. Identifies the counterparty for surveillance/clearing purposes. See CustOrderCapacity (tag 582).
2362	SelfMatchPreventionID	String	N	Self-match prevention token as defined in the order. See Self-Match Prevention. This can be specified in connection-level configuration.
8000	SelfMatchPrevention Strategy	Char	N	Self-match prevention strategy – see Self-Match Prevention for details. See SelfMatchPreventionStrategy (tag 8000) for supported values. This can be specified in connection-level configuration.
1028	ManualOrderIndicator	Boolean	Y	Indicates if the order was initially received manually (as opposed to electronically)
1166	QuoteMsgID	String	N	Refers to QuoteID<117> of a mass quote originally placed the order

<Order Instrument Identity>			Y	
555	NoLegs	NumInGroup	C	Number of instrument legs. Only for multi leg orders
>	*<Instrument Leg Identity>*		C	Instrument of the leg
<Standard Trailer>			Y	

7.18 Order Cancel Reject (MsgType = 9)

Sent by the Small Exchange system to notify the client on rejection of an Order Cancel Request.

Tag	Field Name	Type	Req	Comments
<Standard Header>			Y	MsgType = "9"
1	Account	String	Y	Unique identifier of an account in the system (as specified in the order).
11	ClOrdID	String	Y	ClOrdID identifier of rejected cancel request. See Order Identifiers.
37	OrderID	String	Y	Unique identifier of an order in the Small Exchange system. Maximum length is 20 characters.
39	OrdStatus	Char	Y	Order status value after this rejections. See Order Status (tag 39).
41	OrigClOrdID	String	Y	Order ClOrdID which could not be cancelled/replaced. See Order Identifiers.
60	TransactTime	UTCTimestamp	Y	Time of the reject, expressed in UTC
75	TradeDate	LocalMktDate	N	Business date of the reject
102	CxlRejReason	Int	C	Code to identify the reason for cancel rejection. See Cancel Reject Reason (tag 102).
434	CxlRejResponseTo	Char	Y	Identifies type of request that Cancel Reject is in response to. "1" - order cancel request "2" - order cancel/replace request
58	Text	String	N	Reject reason text. Supports up to 120 characters length.
<Standard Trailer>			Y	

7.19 Business Reject (MsgType = j)

Sent by the Small Exchange system to notify the client of application-level rejection of client request which passed session-level validation but has business validation issues and cannot be reported with a well-formed Execution Report with Rejected status.

Tag	Field Name	Type	Req	Comments
<Standard Header>			Y	MsgType = "j"
45	RefSeqNum	String	Y	MsgSeqNum <34> of rejected message
372	RefMsgType	String	Y	Rejected message MsgType <35>
380	BusinessRejectReason	String	Y	Reject reason code
58	Text	Char	Y	Reject reason text. Supports up to 120 characters length.
<Standard Trailer>			Y	

8 Mass Quote Messages

Mass quoting functionality allows market makers to send a set of two-sided tradable quotes for multiple option instruments within a single message. Each tradable Bid/Ask quote is treated as a DAY Limit Buy/Sell Orders passing the same price limits validations and processed the same way as regular orders.

Mass Quotes management are conducted via the following client-originated messages:

- MassQuote message to submit new or replace existing tradable option quotes
- QuoteCancel message to mass cancel option quotes

The acknowledgements and reports sent back from the Exchange side as results of quotes messages execution:

- Execution Reports (MsgType = 8) sent for each Buy/Sell side of accepted, replaced or cancelled Mass Quote Entries of a Mass Quote
  - New order placed (ExecType = 0)
  - Order is cancelled (ExecType = 4)
  - Trade (ExecType = F)
  - Expired (ExecType = C) – sent for expired Mass Quote Entries after trading session is closed
- Mass Quote Acknowledgement – sent as a common reply to MassQuote or QuoteCancel messages. Sent after all execution reports for quote entries of related processed Mass Quote or Quote Cancel.

8.1 Mass Quote (MsgType = i)

Used to place a set of tradable two-sided quotes for multiple option instruments. Every quote entry being processed cancels existing working quote for QuoteEntryID and places new. If any side of a quote bid or offer is rejected then entire quote is rejected and existing quote is cancelled. There can be only one working Quote Entry for an instrument per single FIX connection. For every placed, cancelled or traded quote entry FIX connection sends corresponding order execution reports for the both buy and sell sides.

Mass Quote messages should meet the requirements otherwise entire message gets rejected:

- Up to 15 Mass Quote Entries can be specified in a message
- All options instruments in Mass Quote Entries of a single message must be of a same underlying futures product (“SM75”, “STIX” and etc)

Tag	Field Name	Type	Req	Comments
<Standard Header>			Y	MsgType = “i”
<Parties>			Y	Operator ID (conveyed via PartyRole = 44) is mandatory.
1	Account	String	Y	Identifier of the mass quotes account. Length up to 10 characters.
117	QuoteID	String	Y	Unique client identifier of the mass quote. Length up to 20 characters.
301	QuoteResponseLevel	Int	Y	Level of response in mass quote acknowledgement to the message: 0 = no acknowledgement 1 = acknowledge erroneous quotes 2 = acknowledge each quote
60	TransactTime	UTC Timestamp	Y	Time this mass quote was initiated by the trading system
528	OrderCapacity	Char	Y	Capacity Origin Code. Identifies the counterparty for surveillance/clearing purposes. See OrderCapacity (tag 528).
582	CustOrderCapacity	Int	Y	Customer Type Indicator Code (CTI) as defined by the NFA. Distinguishes for whom and on what type of account the trade is being placed. See CustOrderCapacity (tag 582).
2362	SelfMatchPreventionID	String	N	Self-match prevention token. To avoid self-match with other firm orders. See Self-Match Prevention. If omitted, default value configured for the connection is used. Both this value and SelfMatchPreventionStrategy (8000) must be defined either in connection-level configuration or in the message.
8000	SelfMatchPrevention Strategy	Char	N	Self-match prevention strategy – see Self-Match Prevention for details. See SelfMatchPreventionStrategy (tag 8000) for supported values. If omitted, default value configured for the connection is used. Both this value and SelfMatchPreventionID (2362) must be

				defined either in connection-level configuration or in the message.
1028	ManualOrderIndicator	Boolean	Y	Indicates if the mass quote was initially received manually (as opposed to electronically)
296	NoQuoteSets	NumInGroup	Y	Must be always “1”
>302	QuoteSetID	String	Y	Must be always “1”
>304	TotNoQuoteEntries	Int	Y	Must be equal to NoQuoteEntries in the message
>295	NoQuoteEntries	NumInGroup	Y	Up to 15 quote entries are supported. Mass Quote with more than 15 entries gets rejected.
>>299	QuoteEntryID	String	Y	Identifier of a tradable quote. Length up to 10 characters.
>>55	Symbol	String	Y	Option symbol of the tradable quote
>>134	BidSize	Qty	N	Bid size of the quote. If not specified or “0” then any existing bid working quantity for the QuoteEntryID gets cancelled.
>>132	BidPx	Price	C	Bid price of the quote, must be specified if BidSize is not 0.
>>135	OfferSize	Qty	N	Offer size of the quote. If not specified or “0” then any existing offer working quantity for the QuoteEntryID gets cancelled.
>>133	OfferPx	Price	C	Offer price of the quote, must be specified if OfferSize is not 0.
<Standard Trailer>			Y	

Mass Quote message example:
BeginString = FIX.4.4
BodyLength = …
MsgType = i
MsgSeqNum = 382
SenderCompID = dx-fix-trade1p
SenderSubID = DX
SendingTime = 20210305-11:45:39.401
TargetCompID = SMALLEX
TargetSubID = TEST
NoPartyIDs = 1
PartyID = mmaker11
PartyIDSource = D
PartyRole = 44
Account = 659848944
QuoteID = 13ab:SM75:qq
QuoteResponseLevel = 2
TransactTime = 20210305-11:45:39.399
OrderCapacity = P
CustOrderCapacity = 2
SelfMatchPreventionID = TOKEN65
SelfMatchPreventionStrategy = B
ManualOrderIndicator = N
NoQuoteSets = 1

QuoteSetID = 1
TotNoQuoteEntries = 2
NoQuoteEntries = 2
QuoteEntryID = 75AUGC71
Symbol = S75E21Q20210820C71.0
BidSize = 2
BidPx = 1.15
OfferSize = 1
OfferPx = 1.33
QuoteEntryID = 75AUGP71
Symbol = S75E21Q20210820P71.0
BidSize = 1
BidPx = 1.54
OfferSize = 1
OfferPx = 1.72
CheckSum = …

8.2 Quote Cancel (MsgType = Z)

Used to mass cancel all working option quotes or for specified underlying futures product sent via the FIX connection.

Tag	Field Name	Type	Req	Comments
<*Standard Header*>			Y	MsgType = "Z"
<*Parties*>			Y	Operator ID (conveyed via PartyRole = 44) is mandatory.
1	Account	String	Y	Identifier of the mass quotes account. Length up to 10 characters.
117	QuoteID	String	Y	Unique client identifier of the quote cancel request. Length up to 20 characters.
298	QuoteCancelType	Int	Y	3 = cancel all option quotes for specified product sent via the FIX connection 4 = cancel all option quotes sent via the FIX connection
60	TransactTime	UTC Timestamp	Y	Time this quote cancel was initiated by the trading system
55	Symbol	String	C	Symbol of options underlying futures product to cancel quotes for. For instance "SM75" or "STIX".
1028	ManualOrderIndicator	Boolean	Y	Indicates if the quote cancel was initially received manually (as opposed to electronically)

Quote Cancel message example:
BeginString = FIX.4.4
BodyLength = …
MsgType = Z
MsgSeqNum = 389
SenderCompID = dx-fix-trade1p
SenderSubID = DX
SendingTime = 20210305-11:45:54.009
TargetCompID = SMALLEX

TargetSubID = TEST
NoPartyIDs = 1
PartyID = mmaker11
PartyIDSource = D
PartyRole = 44
Account = 659848944
QuoteID = ccc:SM75
QuoteCancelType = 3
TransactTime = 20210305-11:45:53.899
Symbol = SM75
ManualOrderIndicator = N
CheckSum = …

8.3 Mass Quote Acknowledgement (MsgType = b)

Is used as a response to Mass Quote and Quote Cancel messages.
Note that QuoteAckStatus = 0 (accepted) acknowledgement does not mean all mass quote entries were accepted, the acknowledgement may contain mixed quote entries with accepted and rejected statuses.

Tag	Field Name	Type	Req	Comments
<*Standard Header*>			Y	MsgType = “b”
<*Parties*>			Y	Operator ID (conveyed via PartyRole = 44) is mandatory.
1	Account	String	Y	Identifier of the mass quotes account. Length up to 10 characters.
117	QuoteID	String	Y	QuoteID<117> of corresponding original mass quote or quote cancel messages.
297	QuoteAckStatus	Int	Y	0 = mass quote message accepted 3 = cancel all option quotes for specified product symbol accepted 4 = cancel all option quotes for the FIX connection accepted 5 = rejected
55	Symbol	String	C	Symbol of options underlying futures product quotes were cancelled for. For instance “SM75” or “STIX”.
300	QuoteRejectReason	Int	N	Mass Quote or Quote Cancel reject reason. See QuoteRejectReason (tag 300).
60	TransactTime	UTC Timestamp	Y	Time this quote cancel was initiated by the trading system
296	NoQuoteSets	NumInGroup	C	Always “1”. Not reported in a reply to quote cancel message. Contains tradable quote entries with a status of acceptance.
>302	QuoteSetID	String	C	Always “1”
>304	TotNoQuoteEntries	Int	C	Equal to NoQuoteEntries
>295	NoQuoteEntries	NumInGroup	C	Up to 15 quote entries are reported
>>299	QuoteEntryID	String	C	Identifier of a tradable quote. Length up to 10 characters.
>>55	Symbol	String	C	Option symbol of the tradable quote.
>>368	QuoteEntryRejectReason	Int	C	Reason code the quote entry was rejected. Empty for accepted quote. See QuoteEntryRejectReason (tag 368).
<*Standard Trailer*>			Y	

Mass Quote Acknowledgement message as a response to Mass Quote example:
BeginString = FIX.4.4
BodyLength = …
MsgType = b
MsgSeqNum = 382
SenderCompID = SMALLEX
SenderSubID = TEST
SendingTime = 20210305-11:45:39.421

TargetCompID = dx-fix-trade1p
TargetSubID = DX
NoPartyIDs = 1
PartyID = mmaker11
PartyIDSource = D
PartyRole = 44
QuoteID = 13ab:SM75:qq
QuoteAckStatus = 0
TransactTime = 20210305-11:45:39.420
NoQuoteSets = 1
QuoteSetID = 1
TotNoQuoteEntries = 2
NoQuoteEntries = 2
QuoteEntryID = 75AUGC71
Symbol = S75E21Q20210820C71.0
QuoteEntryRejectReason = 99
QuoteEntryID = 75AUGP71
Symbol = S75E21Q20210820P71.0
CheckSum = …

Mass Quote Acknowledgement message as a response to Quote Cancel example:
BeginString = FIX.4.4
BodyLength = …
MsgType = b
MsgSeqNum = 382
SenderCompID = SMALLEX
SenderSubID = TEST
SendingTime = 20210305-11:45:39.421
TargetCompID = dx-fix-trade1p
TargetSubID = DX
NoPartyIDs = 1
PartyID = mmaker11
PartyIDSource = D
PartyRole = 44
QuoteID = ccc:SM75
QuoteAckStatus = 3
TransactTime = 20210305-11:45:39.420
CheckSum = …

9 Tag Values

9.1 PartyRole (tag 452)

Value	Description
44	Order Entry Operator ID. Identifier of the operator who placed the order. This is an ‘external’ identifier not assigned by the Exchange.
2101	Subscriber token. If the order is placed by the Exchange subscriber, order messages and execution reports will include a separate ‘party’ with the token value.

9.2 Order Status (tag 39)

Value	Description
0	New
1	Partially filled
2	Filled
3	Done for day
4	Cancelled
6	Pending cancel
8	Rejected
A	Pending new
C	Expired
E	Pending replace

9.3 Order Types (tag 40)

Value	Description
1	Market
2	Limit
3	Stop
4	Stop Limit

9.4 Order Sides (tag 54 and tag 624)

Value	Description
1	Buy side
2	Sell side

9.5 Order Time-in-force (tag 59)

Value	Description
0	Day order
1	Good Till Cancel (GTC)
3	Immediate Or Cancel (IOC)
4	Fill Or Kill (FOK)

9.6 Cancel Reject Reason (tag 102)

Value	Description
0	Too late to cancel
1	Unknown order

9.7 Order Reject Reason (tag 103)

Value	Description
1	Unknown symbol
2	Exchange closed
6	Duplicate order
18	Invalid price increment (submitted price precision exceeds the one supported for the instrument)
99	Other

9.8 Execution Types (tag 150)

Value	Description
0	New, order is placed
3	Done for day
4	Cancelled
5	Replace
8	Rejected
C	Expired
F	Trade
I	Order status

9.9 Session Reject Reason Codes (tag 373)

Value	Description
0	Invalid tag number
1	Required tag missing
3	Undefined Tag
4	Tag specified without a value
5	Value is incorrect (out of range) for this tag
6	Incorrect data format for value
9	CompID problem
99	Other

9.10 Position Effect (tag 77 and tag 564)

The value of these tags indicates the desired direction of a position. The Exchange does not validate the value of these tags.

Value	Description
O	Open
C	Close
D	Default

9.11 CustOrderCapacity (tag 582)

Customer Type Indicator Code as defined by the NFA. Distinguishes for whom and on what type of account the trade is being placed.
See https://www.nfa.futures.org/news/newsNotice.asp?ArticleID=1362.

Value	Description
1	CTI 1: For orders placed by an executing broker for their own account.
2	CTI 2: For orders placed by an executing broker for a firm proprietary account.
3	CTI 3: For orders placed by an executing broker for another broker who also has access to the system.
4	CTI 4: For orders placed by an executing broker on behalf of a customer.

9.12 OrderCapacity (tag 528)

Capacity Origin Code. Identifies the counterparty for surveillance/clearing purposes.

Value	Description
A	Customer's segregated account.
P	Firm's proprietary account.

9.13 SelfMatchPreventionStrategy (tag 8000)

Value	Description
A	Cancel aggressing order
R	Cancel resting order
B	Cancel both aggressing and resting orders

9.14 QuoteRejectReason (tag 300)

Value	Description
99	Other

9.15 QuoteEntryRejectReason (tag 368)

Value	Description
1	Unknown symbol
2	Exchange closed
6	Duplicate quote
8	Invalid price
99	Other